FELCRA Cattle Profiling — Hostinger VPS deployment notes

Oct 11, 2026 · @Sharizan

Deploy the whole system on one Hostinger KVM VPS with Docker: Caddy (HTTPS) in front, the FELCRA app, PostgreSQL 17 and nightly backups. PostgreSQL runs as a Docker container inside the stack, so it needs no separate installation. Production first; staging on the same server is optional. Full reference: `docs/DEPLOYMENT.md` in the repository.

1. What to order

Item	Choice
VPS plan	KVM 2 or higher: 2 vCPU, 8 GB RAM, 100 GB NVMe (KVM 1 with 4 GB works for the pilot only)
Operating system	Ubuntu 24.04 LTS, plain or the template with Docker pre-installed
Data centre	Nearest to Malaysia that Hostinger offers (e.g. Singapore)
Access	SSH key login for the administrator; root password login disabled after set-up
Domain	Production: `profiling.<felcra-domain>`; optional staging: `staging.profiling.<felcra-domain>`
Mail	An SMTP account for the monthly report e-mail (Hostinger mail, FELCRA's mail server or similar)

DNS records (at whoever hosts FELCRA's domain), created before step 6 so HTTPS certificates can be issued:

Type	Name	Value
A	`profiling`	VPS public IPv4
A	`staging.profiling`	VPS public IPv4 (only if staging is wanted)
AAAA	same names	VPS IPv6 (optional)

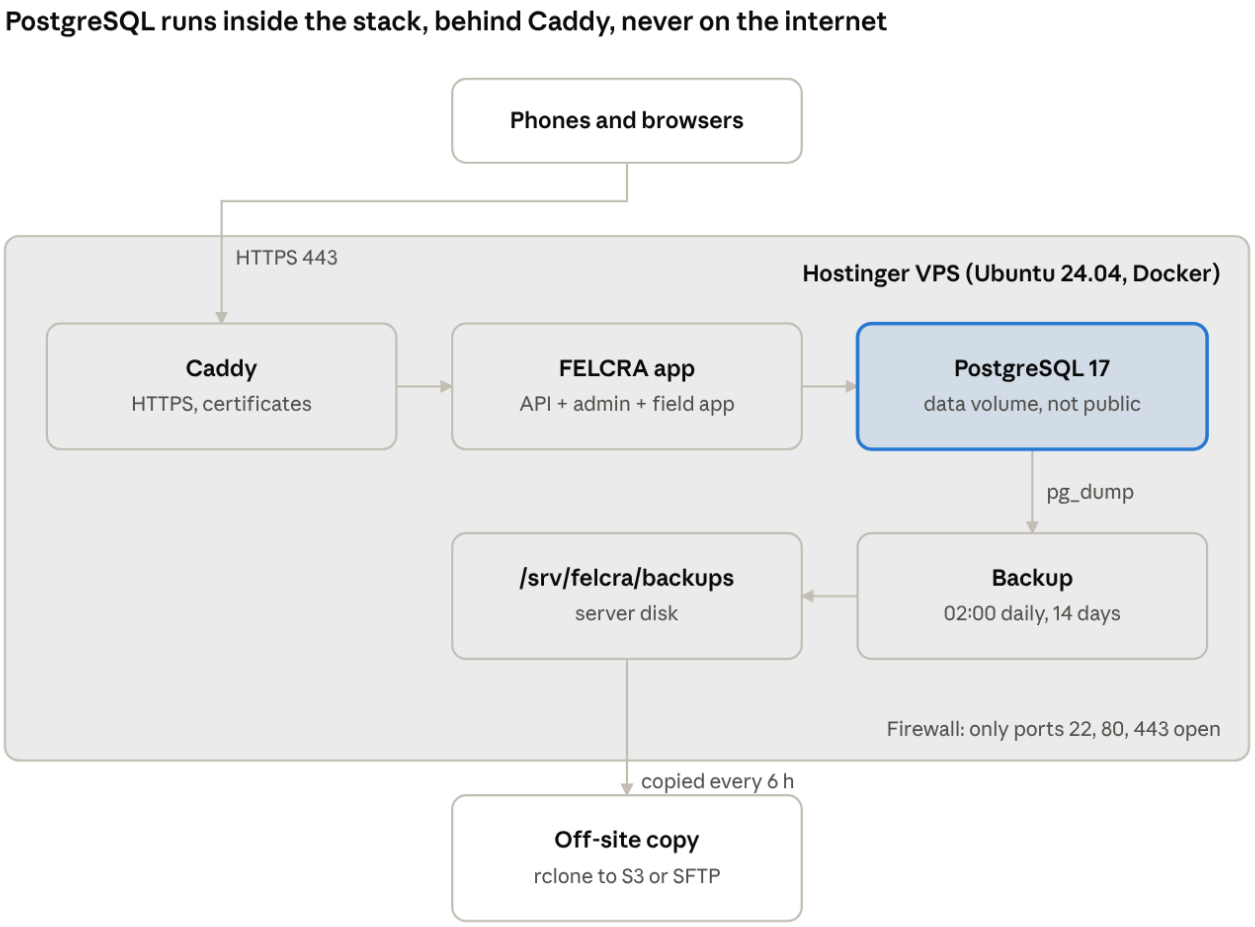
2. What you receive from FELCRA / the developer

Secrets are handed over privately (password manager or in person), never by e-mail or chat, and never stored in the repository.

Item	Used for	Notes
Read access to the GitHub repository `SharizanOsman/felcra_dataprofilling`	Getting the code and deployment files	A deploy key or a read-only access token
Domain names (section 1)	Caddy site addresses	DNS must already point to the VPS
Let's Encrypt contact e-mail	HTTPS certificate notices	e.g. FELCRA IT mailbox
HQ admin e-mail	First administrator account	Created on first start; its password is changed after first sign-in
SMTP host, port, user, password and sender address	Monthly report e-mail	Can be added later; the system runs without it
Pagoh census file `pagoh-census-2026-07.csv`	First data load (section 7)	FELCRA client data: keep it off public places and delete it from the server after import
Backup off-site target	Copy of nightly backups	e.g. an S3-compatible bucket or another server over SFTP

Generated on the server during set-up (not received): PostgreSQL password, JWT signing key, Web Push keys (section 6).

3. At a glance



server layout · 4 containers, 1 off-site copy

Only Caddy is reachable from the internet; it passes requests to the app, which alone talks to PostgreSQL. The backup container dumps the database nightly to the server disk, and an optional copier sends those files off the server.

4. Prepare the server

Only ports 22, 80 and 443 are open to the internet; the database is never exposed. Run as a sudo user, not root.

```
# Updates, firewall, automatic security patches
sudo apt update && sudo apt upgrade -y
sudo apt install -y unattended-upgrades ufw git
sudo ufw allow OpenSSH && sudo ufw allow 80/tcp && sudo ufw allow 443/tcp &&
sudo ufw allow 443/udp
sudo ufw enable

# Docker Engine + Compose plugin (skip if the Ubuntu-with-Docker template was
used)
curl -fsSL https://get.docker.com | sudo sh
sudo usermod -aG docker $USER   # then sign out and in again

# Folders and code
sudo mkdir -p /srv/felcra/backups/production /srv/felcra/backups/staging
sudo chown -R $USER /srv/felcra
git clone https://github.com/SharizanOsman/felcra_dataprofilling.git
/srv/felcra/app
```

If the Hostinger panel firewall is enabled, allow the same three ports there too. Set `PasswordAuthentication no` in `/etc/ssh/sshd_config` once SSH keys work, then `sudo systemctl restart ssh`.

5. PostgreSQL

PostgreSQL 17 is part of the stack: Docker starts it from the official `postgres:17-alpine` image when the app is deployed (section 6). **Do not install PostgreSQL with apt** and do not create the database by hand.

Setting	Value
Image	`postgres:17-alpine` (official)
Database / user	`felcra` / `felcra`
Password	`POSTGRES_PASSWORD` in the stack's env file (generated in section 6)
Data	Docker volume `felcra-production_pgdata` (survives restarts and upgrades)
Network	Reachable only by the app and backup containers; no port published to the host or internet
Tables	Created and upgraded automatically by the app at start-up (database migrations)

Setting	Value
Health	`pg_isready` check; the app waits until the database is healthy

Administrator access, when needed:

```
cd /srv/felcra/app/deploy/app
docker compose -p felcra-production --env-file production.env exec db psql -U
felcra -d felcra
```

Changing `POSTGRES_PASSWORD` after the first start does not change the existing database's password: change it inside PostgreSQL (`ALTER USER felcra PASSWORD '...'`) and in the env file together.

If a separate managed PostgreSQL is preferred instead: version 15 or later, an empty database, and the `ConnectionStrings__Default` line in `deploy/app/compose.yml` pointed at it (the `db` service can then be removed). The container route above is recommended.

6. Deploy

1. **Start the HTTPS proxy (Caddy).** It obtains and renews Let's Encrypt certificates by itself.

   ```
   cd /srv/felcra/app/deploy/proxy
   docker network create felcra-web
   cp sites/production.caddy.example sites/production.caddy   # edit: replace
   the domain
   echo "ACME_EMAIL=it@felcra.com.my" > .env                  # the contact
   e-mail from section 2
   docker compose -p felcra-proxy up -d
   ```

2. **Build the application image** on the server from the code (route A, simplest):

   ```
   cd /srv/felcra/app && git checkout main
   docker build -t felcra-profiling:1.0.0 .
   ```

   Route B, once the developer publishes release images: use `ghcr.io/sharizanosman/felcra-profiling` with a version tag instead, and skip the build.

3. **Create the environment file** for the stack and keep it private:

```
cd /srv/felcra/app/deploy/app
cp env.example production.env && chmod 600 production.env
openssl rand -base64 48        # run twice: POSTGRES_PASSWORD,
JWT_SIGNING_KEY
docker run --rm felcra-profiling:1.0.0 vapid-keys    # PUSH_PUBLIC_KEY /
PUSH_PRIVATE_KEY
```

Variable	Value
`STACK_NAME`	`felcra-production` (must match the proxy site file)
`DOMAIN`	`profiling.<felcra-domain>`
`APP_IMAGE` / `APP_TAG`	`felcra-profiling` / `1.0.0` (route A)
`POSTGRES_PASSWORD`, `JWT_SIGNING_KEY`	Generated above, different values
`ADMIN_EMAIL`, `ADMIN_PASSWORD`	First HQ admin (8+ characters, upper and lower case, a digit)
`SEED_DEMO_DATA`	`false`
`PUSH_PUBLIC_KEY`, `PUSH_PRIVATE_KEY`, `PUSH_SUBJECT`	From `vapid-keys`; subject `mailto:<it e-mail>`. Keep these keys for good
`EMAIL_FROM`, `SMTP_*`, `MONTHLY_REPORTS`	SMTP details; `MONTHLY_REPORTS=true` once mail works
`BACKUP_DIR`	`/srv/felcra/backups/production`

4. **Start the stack** (PostgreSQL, app, backup). The script waits until the app reports healthy:

```
cd /srv/felcra/app/deploy
SKIP_PULL=1 ./deploy.sh production      # route A (local image); route B:
./deploy.sh production 1.0.0
```

5. **Check**: `https://profiling.<felcra-domain>/health` answers `Healthy`, and `/admin/` shows the sign-in page with a valid padlock.

On the very first start the log shows one "Failed executing DbCommand … __ef_migrations_history" error: that is the empty database being checked before its tables are created, and is expected. Staging, if wanted, is the same with `staging` in place of `production` and its own env file and site file.

7. First sign-in and data

Production starts empty apart from the HQ admin account. These steps are done in the web portal, by FELCRA HQ or the developer, not on the server.

1. Sign in at `https://profiling.<felcra-domain>/admin/` with `ADMIN_EMAIL` / `ADMIN_PASSWORD`, then change the password (your name, top right → **Change password**). Then remove `ADMIN_PASSWORD` from `production.env`.
2. **Master data**: projects EPP5 and RP4, breeds, treatments and vaccines (with withdrawal days), feed ingredients.
3. **Farms → Add farm**: PAGOH — FELCRA Berhad Kawasan Pagoh, Segamat, Johor, 319 ha, capacity 200, projects EPP5 and RP4.
4. **Users**: Pagoh farm manager and field officers, with farm PAGOH.
5. **Census → New session** for July 2026 (2–7 Jul, PIC, population on record P 11 / I 61 / AJ 28 / AB 40), then **Import sheet** with `pagoh-census-2026-07.csv`. Expected: 140 animals (P 13, I 60, AJ 31, AB 36), 31 selected.
6. Delete the CSV from wherever it was copied.

8. Backups, monitoring and updates

The `backup` container dumps the database and archives the photos every night at 02:00 Malaysia time into `/srv/felcra/backups/production`, keeping 14 days. Hostinger's own VPS snapshots are a useful extra, not a replacement: they do not give a consistent database copy.

Task	How
Backup now (and test it)	`cd /srv/felcra/app/deploy/app && docker compose -p felcra-production --env-file production.env exec backup /bin/sh /backup.sh now`
Off-site copy (required)	Create `deploy/app/rclone.conf` with a remote named `offsite`, set `OFFSITE_TARGET`, then `docker compose -p felcra-production --env-file production.env --profile offsite up -d offsite`
Restore	`cd /srv/felcra/app/deploy && ./restore.sh production <db-*.dump> [photos-*.tar.gz]` (asks for confirmation; test it once on staging)

Task	How
Uptime monitor	Check `https://profiling.<felcra-domain>/health` every 5 minutes, alert FELCRA IT
Logs	`docker compose -p felcra-production --env-file production.env logs -f app`; daily files in `/app/data/logs` inside the app container (30 days)
Update to a new version	`cd /srv/felcra/app && git pull && docker build -t felcra-profiling:<new> .` then `cd deploy && APP_TAG=<new> SKIP_PULL=1 ./deploy.sh production` — the script backs up first
Rollback	Deploy the previous tag the same way; if the database was changed by the new version, restore the pre-upgrade backup
Disk space	`df -h` monthly; backups and photos grow over time

9. Hand-back checklist

Confirm each item to FELCRA when done:

- [ ] VPS details: IP addresses, plan, data centre, sudo user; root and password SSH login disabled
- [ ] Firewall: only 22, 80 and 443 open (server and Hostinger panel)
- [ ] `https://profiling.<felcra-domain>/health` answers `Healthy`; padlock valid; `/admin/` and `/field/` load
- [ ] `docker ps` shows `caddy`, `app` (healthy), `db` (healthy) and `backup` running
- [ ] PostgreSQL port 5432 not reachable from outside (`nc -zv <ip> 5432` from another machine fails)
- [ ] One manual backup taken, and the files present in `/srv/felcra/backups/production`
- [ ] Off-site copy configured and one copy seen at the destination
- [ ] Uptime monitor on `/health` alerting FELCRA IT
- [ ] `production.env` (mode 600) and its secrets handed to FELCRA IT privately; not left in chat or e-mail
- [ ] SMTP tested: Reports → Monthly report e-mail → **Send me a test e-mail** (if mail details were given)